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Key Personnel Changes/Previous PI:	NOTE: Dan Merfeld took over as PI from Lionel Zupan in Spring 2007; for information prior to FY2007 and for FY2007 report, please see Zupan's task with the same title. For further information, please contact the Task Book Help Desk, taskbook@nasaprs.com		
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Task Description:	Sensory systems often provide ambiguous information. For example, otolith organs measure gravito-inertial force, the sum of gravitational force and inertial force due to linear acceleration. According to Einstein's Equivalence Principle, no set of linear accelerometers alone can distinguish gravitational force (which changes with head orientation during a head tilt) from inertial force (which changes with linear acceleration of the head). Therefore, the Central Nervous System (CNS) must use other sensory cues to distinguish tilt from translation. For example, the CNS can use rotational cues provided by the semicircular canals and vision. To investigate the influence of rotational cues on the neural processing of tilt and translation in humans, we propose to develop a countermeasure to adapt and/or pre-adapt astronaut canal-otolith interactions to any gravitational environment (Earth, Mars, …) by providing correlated sensory cues between the canals and the otoliths. The countermeasure we propose to develop, often referred to as Artificial Gravity, consists of centrifugation combined with intermittent motion. We will measure behavioral (eye movements such as vestibulo-ocular reflexes) as well as psychophysical responses (orientation estimation) during tilt alone, translation alone or combined tilt-translation motion in either a 1-G or hypergravic environment. Fourier analysis and modeling tools will also be used to investigate the neural processing between rotational and gravito-inertial cues. This Task addresses the following specific aims (SA) using human subjects: SA I. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses). This specific aim addresses Critical Question 9.01 since SA 1 will provide baseline data for humans to evaluate if Artificial Gravity combined with intermittent motion is a valid countermeasure (SA 2). We propose a ground study to measure human responses (perceptual tilt and eye movements) during combined canal-otolith stimulation on Earth, in a 1-G environment. Our goal is to provide a perceptual measure of tilt with simultaneous eye movement recording to comprehensively investigate how the human Central Nervous System distinguishes tilt from translation. Subjects will be tested using protocols in which they passively a) tilt, b) translate, or c) tilt & translate simultaneously. SA II. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses) by measuring adaptive changes induced by exposure to a hyper-G environment. This specific aim addresses both Critical Questions (CQ) 9.01 and 9.04 by evaluating if Artificial Gravity combined with intermittent motion is a valid countermeasure (CQ 9.01) and by measuring adaptive changes induced by exposure to altered-gravity (CQ 9.04). We propose to measure human perception of orientation and eye movement responses during exposure to artificial gravity. To induce adaptation, we propose to utilize intermittent motion during centrifugation: passive roll tilt in a supine orientation will dynamically stimulate both semicircular canals and otolith organs, such that both sensory cues are correlated. (The motion envelope is in an Earth-horizontal plane.) For Specific Aim #1, we expect the human perceptual responses to be somewhat consistent with those previously measured in monkeys by Angelaki and Colleagues (1999), with the canal cue essential for accurate estimation of gravity and linear acceleration, but we also expect differences in eye responses as shown by our preliminary data with a significant horizontal eye response to roll tilt at high frequencies. The GIF resolution hypothesis states that the CNS estimates gravity and linear acceleration such that their mathematical difference matches the graviceptor measurement of the GIF. Therefore, for Specific Aim #2, based on the GIF resolution hypothesis, we expect that: a) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1-G linear acceleration (Expt 2A) to be similar as for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment since both dynamic canal and otolith cues are the exact same (except for a static naso-occipital 1-G gravitational acceleration in 2A): such a result would validate artificial gravity combined with intermittent motion as a countermeasure. The responses may not be exactly similar during Trial 1 but we expect the CNS to have adapted by Trial 4 back to a response similar to the response in Expt. 1 (1-G environment). b) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1.2-G (or 1.4 G) linear acceleration (Expt 2B) to be originally (Trial 1) different from responses for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment: for example, we expect a larger vertical VOR response (Trial 1). Then we expect the CNS to adapt to this Hyper-G environment: for example, the vertical VOR response should decrease while the horizontal VOR response increases as the CNS adapts to the hyper-G environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since all proposed experiments are ground studies, the Earth Benefits should be straight forward. Experimental data shoud provide a better insight into how the Central Nervous System separates the graviceptor measurements of gravito-inertial force into estimates of gravity and linear acceleration, or in other words, into estimates of tilt and translation. This central process is essential to a number of everyday task such as visual stabilization or postural control and its understanding is therefore essential to the understanding of these many central processes. In addition, these ground studies should also provide insights on how the Central Nervous System adapts this separation process when humans experience a new gravitational environment. Understanding this process should not only help train and condition astronauts to microgravity or to a return to Earth-gravity after a long stay in microgravity, but it should also help us understand the rehabilitation process in vestibular patients and therefore help us design better methods of rehabilitation.
	Final Report for NNJHB01G P.I. Daniel M. Merfeld, Ph.D. Massachusetts Eye and Ear Infirmary "The influence of rotational cues on human tilt and translation reponses" 5/15/04 – 8/31/09 Scientific goals/objectives: Scientifc goals were to understand how our brain combines sensory cues to yield tilt and translation sensations. Final Progress Report: The progress is best summarized by a published peer-reviewed paper funded by this grant and a second manuscript that has been submitted for peer review. Immediately below, we provide the abstracts from both of these papers. For completeness, we will append the proof of the full peer-reviewed paper and the full submitted manuscript.

Task Progress:	The abstract for the peer-reviewed paper follows: Using a variable-radius roll swing motion paradigm, we examined the influence of inter-aural (y-axis) and dorso-ventral, (z-axis) force modulation on perceived tilt and translation by measuring perception of horizontal translation, roll tilt, and distance from center of rotation (radius) at 0.45 and 0.8 Hz using standard magnitude estimation techniques (primarily verbal reports) in darkness. Results show that motion perception was significantly influenced by both y- and z-axis forces. During constant radius trials, subjects' perceptions of tilt and translation were generally almost veridical. By selectively pairing radius (1.22 m and 0.38 m) and frequency (0.45 and 0.8 Hz, respectively), the y-axis acceleration could be tailored in opposition to gravity so that the combined y-axis gravitoinertial force (GIF) variation at the subject's ears was reduced to about 0.035 m/s2 – in effect, the y-axis GIF was "nulled" below putative perceptual threshold levels. With y-axis force nulling, subjects overestimated their tilt angle and underestimated their horizontal translation and radius. For some trials a radial linear acceleration at twice the tilt frequency (0.25 m/s2 at 0.9 Hz, 0.13 m/s2 at 1.6 Hz) was simultaneously applied to reduce the z-axis force variations caused by centripetal acceleration and by changes in the z-axis component of gravity during tilt. For other trials, the phase of this radial linear acceleration was altered in order to double the magnitude of the z-axis force variations. Z-axis force nulling further increased the perceived tilt angle and further decreased perceived horizontal translation and radius relative to the y-axis nulling trials, while z-axis force doubling had the opposite effect. Subject reports were remarkably geometrically consistent; an observer model-based analysis suggests that perception was influenced by their knowledge of swing geometry. The abstract for the submitted manuscript, which we entitled "Perceived tilt and translation during variable-radius swing motion with congruent or conflicting visual and vestibular cues", follows: We examined the influence of a realistic dynamic visual scene on the motion perception of subjects undergoing sinusoidal (0.45 Hz) roll swing motion at different radii. The visual scene, presented on a flatscreen monitor with a 40° field of view and viewed monocularly, showed visual motion with a rotation radius either the same as or different from the subject's actual rotation radius. There were 3 categories of trials: 1. Trials in the dark with no visual cues; 2. Trials where the projected visual scene matched the actual motion; and 3. Trials where the projected visual scene showed swing motion at a different radius. Subjects verbally reported their perceptions using standard magnitude estimation techniques. Trained subjects reported motion perceptions that were geometrically consistent with radii intermediate between the radii of the visual scene and the actual movement. The visual scene exerted a stronger influence than the actual movement at this frequency. Subjects were generally unable to detect the cue conflicts or judge their own visual-vestibular biases. Awards, publications, presentations: One peer-reviewed paper has been published. The citation is: Rader, A. A., Oman, C. M. & Merfeld, D. M. Motion perception during variable-radius swing motion in darkness. Journal of neurophysiology, doi:00116.2009 [pii] 10.1152/jn.00116.2009 (2009). A second manuscript has been submitted to the Journal of Neurophysiology for publication. The title of this manuscript is: "Perceived tilt and translation during variable-radius swing motion with congruent or conflicting visual and vestibular cues." New Technology: The eccentric rotator purchased by this grant has been installed and is now fully operational. Both the hardware and software seem to working extremely well, so we consider this new technology fully operational. Future Research Plans: This project is no longer funded by NASA. We do have NIH funding and anticipate pursuing our interests in sensory integration, in general, and canal-otolith interactions in particular via this funding source.

Bibliography Type:	Description: (Last Updated: 04/05/2011)
Articles in Peer-reviewed Journals	Rader AA, Oman CM, Merfeld DM. "Motion perception during variable-radius swing motion in darkness." J Neurophysiol. 2009 Oct;102(4):2232-44. Epub 2009 Jul 22. PMID: 19625542 , Oct-2009
Articles in Peer-reviewed Journals	Zupan LH, Merfeld DM. "Interaural self-motion linear velocity thresholds are shifted by roll vection." Exp Brain Res. 2008 Dec;191(4):505-11. Epub 2008 Oct 9. PMID: 18843487 , Dec-2008
Articles in Peer-reviewed Journals	Rader AA, Oman CM, Merfeld DM. "Perceived tilt and translation during variable-radius swing motion with congruent or conflicting visual and vestibular cues." Exp Brain Res. 2011 Apr;210(2):173-84. Epub 2011 Mar 19. PMID: 21424258 , Apr-2011